Shiv Nadar University - Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 14/10/2026 (Week 11) **Continuous Lab Evaluation-11** II Yr/ 3rd Sem B.Tech CSE(IoT)A&B

Allotted Marks: 10 (Hardware Implementation + Code Efficiency + Viva-Voce)

Topics: CubeMX + Keil MDK + **STM32 Nucleo Development Board**

STM32 HAL libraries are allowed

STM32F401RE Cortex-M4: GPIO and User Input Programming

Prerequisite :

1. Observe the on-board oscillator frequency.
2. Check the STM32 Controller on the board and understand the specifications like clock frequency, GPIO Group, #UARTs, #I2Cs, #SPIs any other worth mentioning.
3. Decide on the Clock configuration settings in Cube-MX.
4. Refer to the Nucleo Board STM32F401RE/RB Pinout.
5. https://www.st.com/resource/en/user_manual/um1724-stm32-nucleo64-boards-mb1136-stmicroelectronics.pdf

Task 1 — On-board LED Bring-Up

Develop and program an STM32 application to verify the NUCLEO-F401RE hardware and GPIO output configuration by blinking the on-board LED at a fixed interval.

Task 2 — On-board Switch Control with State Remembrance

Modify the application so that the on-board push button controls the LED. The LED shall toggle its state whenever a valid button press is detected and shall remember its previous state after the button is released.

Task 3 — Short-Press and Long-Press Detection

Extend the application to distinguish between a short button press and a long button press. The STM32 shall perform different actions depending on the duration of the button press.